

Franciszek Wiercigroch

In a few sentences, explain the topic of your capstone project. The write-up need not be technical and should be accessible to other data science minors (even if they have not had the same course background as you).

Also, included in the topic, please remind me the name of your faculty advisor.

For the first draft, clearly and precisely spell out both the **data** aspect of your project as well as the **advanced data-analytic method** you will incorporate. Eventually the capstone topic draft will turn into the introduction for your final write-up. At that point (but not for the first draft), it will include an explanation of the scope of the project, and its value, as well as a short introduction to the sources. It will also include definitions of technical terms needed to understand your project.

My capstone project this semester will be focusing on working with the Pomona College faculty and staff commute data from fiscal year 2026. The goal of the project is to decrease faculty and staff commute emissions in their chosen mode of transportation through sustainable transportation options based on the projected goals Transportation page of the 2026 Annual Save Report. This will be done in conjunction with the Pomona College sustainability office, with my advisor being Alexis Reyes.

The dataset contains faculty and staff commuting information with columns detailing their home zip code, one-way distance to work, reported travel mode for each weekday, amongst other basic summary statistics. This will be the primary data used to analyze commuting patterns to see where commuting modes could potentially be switched, such as overreliance on driving and seeing where users can be more aware of the environmental impacts of their commute method with potential for carpooling or public transit.

The advanced data-analytics method I will incorporate will be a combination of advanced data visualization through a dashboard and me scraping multiple websites to get public transit data for the purposes of the dashboard. I will likely have to conduct some form of geospatial analysis to identify the geographic clusters of employees who have similar commute patterns. I will be integrating transit routes through Metrolink and other SoCal public transit options to integrate within the dashboard.